

Disproof of Conjecture 00000003831: a seminormal $\mathfrak{gl}_3$ crystal whose character is $s_{21} - s_{111}$

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Abstract

Conjecture 00000003831 asserts that for every finite crystal B the Schur expansion of the Weyl symmetrization of the character $\text{ch}(B)$ has nonnegative coefficients, and that for $B = B(\lambda) \otimes B(\mu)$ these coefficients are the Littlewood–Richardson coefficients. The first clause is false. We exhibit a connected 7-vertex *seminormal* $\mathfrak{gl}_3$ crystal B_7 , obtained from $B(2, 1, 0)$ by deleting one of its two vertices of weight $(1, 1, 1)$ and re-gluing the strings through the other one. It satisfies all of Kashiwara’s crystal axioms, ε_i, φ_i are the string lengths, and

$$\text{ch}(B_7) = m_{21} + m_{111} = s_{21} - s_{111}.$$

The character is symmetric, so the Weyl symmetrizer $J(x^\rho f)/J(x^\rho)$ fixes it and the coefficient of s_{111} is -1 ; the orbit sum gives $6s_{21} - 6s_{111}$ and the orbit average gives $s_{21} - s_{111}$. Seven vertices is the minimum for a seminormal $\mathfrak{gl}_3$ counterexample, and seminormal $\mathfrak{gl}_2$ crystals never give one. Clause 1 is true for *normal* (Stembridge) crystals, so it can only be rescued by reading “finite crystal” as “normal crystal”. The refutation is formalised in Lean 4 (core library only): crystals, Kashiwara’s axioms, seminormality, characters, Schur polynomials (from tableaux), the antisymmetrizer and the Weyl symmetrization are defined from scratch. An independent Python script rechecks everything and performs the minimality searches.

1 The claim

The conjecture (English version): “*Definition: the crystal character $\text{ch}(B)$ denotes the generating function of vertex weights. Conjecture: for any finite crystal, the coefficients of the Schur expansion of the Weyl symmetrization of $\text{ch}(B)$ are nonnegative, and for $B = B(\lambda) \otimes B(\mu)$ these coefficients are exactly the classical Littlewood–Richardson coefficients.*”

It is a conjunction of two clauses:

- (C1) for every finite crystal B , the Schur expansion of the Weyl symmetrization of $\text{ch}(B)$ has nonnegative coefficients;
- (C2) for $B = B(\lambda) \otimes B(\mu)$ these coefficients are the LR coefficients $c_{\lambda\mu}^\nu$.

We refute (C1); one false clause makes the conjunction false. Under the Weyl-symmetrizer reading (C2) is *true*: $\text{ch}(B(\lambda) \otimes B(\mu)) = s_\lambda s_\mu$ is symmetric, so it is fixed by the symmetrizer, and $s_\lambda s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$. The disproof rests entirely on (C1).

2 Definitions

Crystals. We work with $\mathfrak{gl}_3$: weight lattice $P = \mathbb{Z}^3$ with basis $\epsilon_1, \epsilon_2, \epsilon_3$, simple roots $\alpha_1 = \epsilon_1 - \epsilon_2$, $\alpha_2 = \epsilon_2 - \epsilon_3$, and coroots acting by $\langle h_1, w \rangle = w_1 - w_2$, $\langle h_2, w \rangle = w_2 - w_3$. Following Kashiwara [2, 3] and Bump–Schilling [1, Ch. 2], a *crystal* is a set B with maps $\text{wt}: B \rightarrow P$, $\varepsilon_i, \varphi_i: B \rightarrow \mathbb{Z} \sqcup \{-\infty\}$ and $\tilde{e}_i, \tilde{f}_i: B \rightarrow B \sqcup \{0\}$ ($i = 1, 2$) such that

1. $\varphi_i(b) = \varepsilon_i(b) + \langle h_i, \text{wt}(b) \rangle$;
2. $\tilde{f}_i b = b'$ if and only if $\tilde{e}_i b' = b$;
3. if $\tilde{e}_i b \in B$ then $\text{wt}(\tilde{e}_i b) = \text{wt}(b) + \alpha_i$, $\varepsilon_i(\tilde{e}_i b) = \varepsilon_i(b) - 1$, $\varphi_i(\tilde{e}_i b) = \varphi_i(b) + 1$;
4. if $\tilde{f}_i b \in B$ then $\text{wt}(\tilde{f}_i b) = \text{wt}(b) - \alpha_i$, $\varepsilon_i(\tilde{f}_i b) = \varepsilon_i(b) + 1$, $\varphi_i(\tilde{f}_i b) = \varphi_i(b) - 1$;
5. if $\varphi_i(b) = -\infty$ then $\tilde{e}_i b = \tilde{f}_i b = 0$.

A crystal is *seminormal* if $\varepsilon_i(b) = \max\{k \geq 0 : \tilde{e}_i^k b \neq 0\}$ and $\varphi_i(b) = \max\{k \geq 0 : \tilde{f}_i^k b \neq 0\}$ for all i, b ; it is *normal* if it is (isomorphic to) the crystal of an integrable $U_q(\mathfrak{gl}_3)$ -module, equivalently a disjoint union of highest-weight crystals $B(\lambda)$ (for simply-laced types this is equivalent to Stembridge’s local axioms [4]). In these references seminormality and normality are *extra* conditions; “crystal” alone means the abstract notion above.

Characters and Schur polynomials. For a finite crystal, $\text{ch}(B) = \sum_{b \in B} x^{\text{wt}(b)} \in \mathbb{Z}[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}]$. The Weyl group S_3 permutes the variables. Let $J(f) = \sum_{w \in S_3} \text{sgn}(w) w(f)$, $\rho = (2, 1, 0)$ and $a_\mu = J(x^\mu)$. For a dominant weight λ ($\lambda_1 \geq \lambda_2 \geq \lambda_3$, integers) the Schur polynomial is $s_\lambda = a_{\lambda+\rho}/a_\rho$; for a partition λ it equals $\sum_T x^T$ over semistandard Young tableaux of shape λ with entries in $\{1, 2, 3\}$, and $s_{\lambda+(k,k,k)} = (x_1 x_2 x_3)^k s_\lambda$.

Lemma 1 (uniqueness of Schur expansions). *The s_λ , λ dominant, are linearly independent, and a symmetric Laurent polynomial f equals $\sum_\lambda c_\lambda s_\lambda$ with $c_\lambda = [x^{\lambda+\rho}] J(x^\rho f)$.*

Proof. If f is symmetric then $J(x^\rho f) = f a_\rho$, an antisymmetric Laurent polynomial. An antisymmetric Laurent polynomial is determined by its coefficients on the strictly decreasing exponents $\mu_1 > \mu_2 > \mu_3$ (every exponent with two equal entries has coefficient 0, and the others are $\pm$ a strictly decreasing one), so $J(x^\rho f) = \sum_\lambda c_\lambda a_{\lambda+\rho}$ with c_λ as stated. Dividing by a_ρ gives the expansion. If $\sum c_\lambda s_\lambda = 0$ then $\sum c_\lambda a_{\lambda+\rho} = 0$, and the a_μ with distinct strictly decreasing μ have disjoint supports, so all $c_\lambda = 0$. $\square$

Readings of “Weyl symmetrization”. For a Laurent polynomial f :

- (W) the Weyl symmetrizer $\pi(f) = J(x^\rho f)/J(x^\rho)$ (the Weyl character formula operator, equal to the Demazure operator $\pi_{w_0} = \pi_1 \pi_2 \pi_1$); its Schur coefficients are $[x^{\lambda+\rho}] J(x^\rho f)$ for any f ;
- (O) the orbit sum $\sum_{w \in S_3} w(f)$;
- (A) the orbit average $\frac{1}{6} \sum_{w \in S_3} w(f)$.

If f is symmetric then $\pi(f) = f$ (because $J(x^\rho f) = f J(x^\rho)$), the orbit sum is $6f$ and the average is f .

3 The counterexample

The crystal B_7 has seven vertices

$$A(2, 1, 0), B(1, 2, 0), C(2, 0, 1), D(0, 2, 1), E(1, 0, 2), F(0, 1, 2), Z(1, 1, 1)$$

(weights in brackets), and the edges

$$\tilde{f}_1 : A \rightarrow B, \quad C \rightarrow Z \rightarrow D, \quad E \rightarrow F; \quad \tilde{f}_2 : A \rightarrow C, \quad B \rightarrow Z \rightarrow E, \quad D \rightarrow F,$$

with $\tilde{e}_i$ the inverse partial maps and $\tilde{e}_i, \tilde{f}_i = 0$ elsewhere. The values of ε_i, φ_i are the string lengths:

	A	B	C	D	E	F	Z
wt	(2, 1, 0)	(1, 2, 0)	(2, 0, 1)	(0, 2, 1)	(1, 0, 2)	(0, 1, 2)	(1, 1, 1)
$(\varepsilon_1, \varphi_1)$	(0, 1)	(1, 0)	(0, 2)	(2, 0)	(0, 1)	(1, 0)	(1, 1)
$(\varepsilon_2, \varphi_2)$	(0, 1)	(0, 2)	(1, 0)	(0, 1)	(2, 0)	(1, 0)	(1, 1)

The crystal $B(2, 1, 0)$ has the eight tableaux $11/2, 12/2, 11/3, 12/3, 13/2, 13/3, 22/3, 23/3$ (written first row/second row), with $\tilde{f}_1$ -edges $11/2 \rightarrow 12/2, 11/3 \rightarrow 12/3 \rightarrow 22/3, 13/3 \rightarrow 23/3$ and $\tilde{f}_2$ -edges $11/2 \rightarrow 11/3, 12/2 \rightarrow 13/2 \rightarrow 13/3, 22/3 \rightarrow 23/3$. The two vertices of weight (1, 1, 1) are $12/3$ (on a 1-string of length 2, alone in colour 2) and $13/2$ (alone in colour 1, on a 2-string of length 2). Identifying them to a single vertex Z keeps all eight edges and gives exactly B_7 .

Proposition 2. B_7 is a connected seminormal crystal (with all ε_i, φ_i finite).

Proof. Each $\tilde{f}_i$ is injective on its domain, so $\tilde{e}_i$ is a well-defined inverse and axiom 2 holds. Each edge $u \rightarrow v$ of colour i has $\text{wt}(u) - \text{wt}(v) = \alpha_i$ (check the four edges of each colour, e.g. $C \rightarrow Z$: $(2, 0, 1) - (1, 1, 1) = (1, -1, 0) = \alpha_1$, $Z \rightarrow E$: $(1, 1, 1) - (1, 0, 2) = (0, 1, -1) = \alpha_2$). The i -strings are $A \rightarrow B, C \rightarrow Z \rightarrow D, E \rightarrow F$ for $i = 1$ and $A \rightarrow C, B \rightarrow Z \rightarrow E, D \rightarrow F$ for $i = 2$; every vertex lies on exactly one string of each colour. Setting $\varepsilon_i(b)$ (resp. $\varphi_i(b)$) to the number of steps from b to the top (resp. bottom) of its string gives the table above, makes B_7 seminormal by definition, and gives axioms 3 and 4 automatically. Axiom 1 is a direct check: for a string $b_0 \rightarrow \dots \rightarrow b_\ell$ of colour i we have $\langle h_i, \text{wt}(b_j) \rangle = \langle h_i, \text{wt}(b_0) \rangle - 2j$, and from the table $\langle h_i, \text{wt}(b_0) \rangle = \ell$ in all six strings (for instance $\langle h_1, \text{wt}(C) \rangle = 2 - 0 = 2$, $\langle h_2, \text{wt}(B) \rangle = 2 - 0 = 2$, $\langle h_1, \text{wt}(A) \rangle = 1$). Hence $\varphi_i(b_j) - \varepsilon_i(b_j) = (\ell - j) - j = \langle h_i, \text{wt}(b_j) \rangle$. Axiom 5 is void. Connectedness: every vertex is joined to A . $\square$

Proposition 3. $\text{ch}(B_7) = s_{(2,1,0)} - s_{(1,1,1)}$, and this is the unique Schur expansion.

Proof. The weights of B_7 are the six permutations of (2, 1, 0) and (1, 1, 1), each once, so $\text{ch}(B_7) = m_{21} + m_{111}$, which is symmetric. The semistandard tableaux of shape (2, 1) with entries ≤ 3 are the eight listed above, so $s_{21} = m_{21} + 2m_{111}$, and $s_{111} = x_1 x_2 x_3 = m_{111}$. Hence $\text{ch}(B_7) = s_{21} - s_{111}$. Uniqueness is Lemma 1; in degree 3 it is also visible directly, since the coefficients of $x^{300}, x^{210}, x^{111}$ in $s_{300}, s_{210}, s_{111}$ form the unitriangular matrix $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix}$. $\square$

Theorem 4. Clause (C1) fails for B_7 under each of the readings (W), (O), (A):

$$\pi(\text{ch } B_7) = s_{21} - s_{111}, \quad \sum_w w(\text{ch } B_7) = 6s_{21} - 6s_{111}, \quad \frac{1}{6} \sum_w w(\text{ch } B_7) = s_{21} - s_{111}.$$

The coefficient of s_{111} is $-1, -6, -1$ respectively. Consequently the conjecture is false.

Proof. $\text{ch } B_7$ is symmetric, so by the remark after the readings the three symmetrizations are $f, 6f, f$ with $f = \text{ch } B_7$; Proposition 3 and Lemma 1 give the expansions. Explicitly, $J(x^\rho \text{ch } B_7) = a_{(4,2,0)} - a_{(3,2,1)}$, so $[x^{(1,1,1)+\rho}]J(x^\rho \text{ch } B_7) = -1$. $\square$

4 What is true, and the boundary of the counterexample

Normal crystals. If B is normal then $B \cong \bigsqcup_k B(\lambda^{(k)})$ and $\text{ch } B = \sum_k s_{\lambda^{(k)}}$ (the character of $B(\lambda)$ is the irreducible character s_λ), which is symmetric and Schur-nonnegative; all three readings then give nonnegative coefficients. In particular (C1) holds for $B(\lambda) \otimes B(\mu)$, and (C2) holds under (W) and (A); under (O) the coefficients are $6c_{\lambda\mu}'$, so (C2) fails under (O). Our B_7 is not normal: a normal crystal has a Schur-positive character. So the only reading under which (C1) survives is “finite crystal” = “normal crystal”. The literal statement says “any finite crystal”, and in the standard references [2, 3, 1] “crystal” does not include normality (or even seminormality); B_7 is moreover seminormal, the strongest condition short of normality that is usually attached to the word.

Characters of seminormal crystals.

Lemma 5. *Let B be a finite seminormal $\mathfrak{gl}_r$ crystal and $m(\nu)$ the number of vertices of weight ν . Then m is S_r -invariant, and for every i and every ν with $\langle h_i, \nu \rangle \geq 0$ we have $m(\nu + \alpha_i) \leq m(\nu)$ (the string condition). Conversely every finitely supported $m \geq 0$ with these two properties is the weight multiplicity function of a finite seminormal crystal.*

Proof. Seminormality makes every vertex lie on a unique i -string $b_0 \rightarrow \cdots \rightarrow b_\ell$ with $\varepsilon_i(b_0) = 0$, $\varphi_i(b_0) = \ell$, so $\langle h_i, \text{wt } b_0 \rangle = \ell$ and the weights $\text{wt } b_0 - j\alpha_i$, $0 \leq j \leq \ell$, are permuted by the reflection s_i . Hence m is s_i -invariant for all i , i.e. S_r -invariant. If $\langle h_i, \nu \rangle = k \geq 0$, a string through $\nu + \alpha_i$ contains $s_i(\nu + \alpha_i) = \nu - (k+1)\alpha_i$ and is contiguous, so it contains ν ; this gives $m(\nu + \alpha_i) \leq m(\nu)$. Conversely, for each colour separately, cut every line $\nu + \mathbb{Z}\alpha_i$ into strings by the usual layering (the multiplicities along the line are symmetric about the reflecting point and weakly decrease away from it), and define $\tilde{f}_i, \tilde{e}_i, \varepsilon_i, \varphi_i$ along these strings. The axioms only involve one colour at a time, so the colours do not interact. $\square$

Proposition 6 ($\mathfrak{gl}_2$). *Every finite seminormal $\mathfrak{gl}_2$ (or $\mathfrak{sl}_2$) crystal has a Schur-positive character; so no counterexample exists for $\mathfrak{gl}_2$ among seminormal crystals.*

Proof. There is one colour; B is the disjoint union of its strings, and a string with top weight (a, c) has $\ell = a - c \geq 0$ and character $\sum_{j=0}^{\ell} x_1^{a-j} x_2^{c+j} = s_{(a,c)}$. $\square$

For $\mathfrak{gl}_3$ each colour still decomposes $\text{ch } B$ into $\mathfrak{sl}_2$ -characters, but the two decompositions need not come from a common $\mathfrak{gl}_3$ -structure; B_7 exploits exactly this.

Proposition 7 (minimality). *Every finite seminormal $\mathfrak{gl}_3$ crystal with at most 6 vertices has a Schur-positive character. Hence 7 is the minimum number of vertices of a seminormal $\mathfrak{gl}_3$ counterexample to (C1).*

Proof. Edges preserve the degree $w_1 + w_2 + w_3$, and the Schur expansion is compatible with the decomposition by degree, so a minimal counterexample lives in one degree d ; multiplying by $(x_1 x_2 x_3)^k$ maps s_λ to $s_{\lambda + (k,k,k)}$ and preserves everything, so we may take $d \in \{0, 1, 2\}$. If a dominant λ occurs, $p = \lambda_1 - \lambda_2$, then by Lemma 5 the $p+1$ distinct weights $\lambda - j\alpha_1$ ($0 \leq j \leq p$) all occur, so $p \leq N-1$, and likewise $\lambda_2 - \lambda_3 \leq N-1$, where N is the number of vertices. This leaves a finite list of multiplicity functions; `verify.py` enumerates all of them for $N \leq 6$, keeps those satisfying the conditions of Lemma 5 (necessary conditions suffice here) and finds exactly 12, namely $k s_{000}$ ($1 \leq k \leq 6$), s_{100} , $2s_{100}$, s_{110} , $2s_{110}$, s_{200} and $s_{(1,1,-1)}$, all Schur-positive. For $N = 7$ there is exactly one non-positive admissible character up to the shift, $m_{(1,0,-1)} + m_{000} = s_{(1,0,-1)} - s_{000}$, which is $\text{ch}(B_7)$ shifted by $(x_1 x_2 x_3)^{-1}$; the script also builds a seminormal crystal for it and rechecks the axioms. $\square$

Remark 8 (a cheaper abstract example). Without seminormality, one vertex suffices: the $\mathfrak{gl}_2$ crystal $\{b\}$ with $\text{wt}(b) = (-1, 2)$, $\varepsilon(b) = 3$, $\varphi(b) = 0$, $\tilde{e}b = \tilde{f}b = 0$ satisfies Kashiwara’s axioms ($\varphi - \varepsilon = -3 = \langle h, \text{wt } b \rangle$). Then $J(x^\rho x_1^{-1} x_2^2) = a_{(0,2)} = -a_{(2,0)}$, so the Weyl symmetrization is $-s_{(1,0)}$, and the orbit sum is $s_{(2,-1)} - s_{(1,0)}$. We do not rely on this example: it is far from the spirit of the conjecture, while B_7 is seminormal, connected and differs from $B(2, 1, 0)$ only by one vertex.

5 Readings covered

- *Crystal*: abstract Kashiwara crystals and seminormal crystals (B_7 is seminormal). Only “normal crystal” rescues (C1).
- *Weyl symmetrization*: the symmetrizer $J(x^\rho f)/J(x^\rho)$ (Demazure operator π_{w_0}), its bialternant coefficients $[x^{\lambda+\rho}]J(x^\rho f)$, the orbit sum and the orbit average all give a negative coefficient. Since $\text{ch } B_7$ is symmetric, any symmetrization that fixes symmetric functions, or multiplies them by a positive constant, does.
- $\mathfrak{gl}_3$ versus $\mathfrak{sl}_3$: projecting the weights to the $\mathfrak{sl}_3$ weight lattice turns s_λ into the irreducible character $\chi_{\bar{\lambda}}$, and $\text{ch } B_7 = \chi_{(1,1)} - \chi_{(0,0)}$ (in fundamental-weight coordinates); the coefficient of the trivial character is -1 .
- *Polynomial versus Laurent*: all weights of B_7 are nonnegative, so $\text{ch } B_7$ is an honest polynomial and the expansion is in ordinary Schur polynomials.

6 Machine verification

Lean 4 (`lean4/Main.lean`, Lean 4.19.0, core library only, no `sorry`, no `native_decide`; the main theorems use only `propext` and `Quot.sound`, some lemmas also `Classical.choice`). Everything is defined from scratch:

- `Crystal n`: $\text{wt}: \text{Fin } n \rightarrow \mathbb{Z}^3$, $\tilde{e}_i, \tilde{f}_i: \text{Fin } n \rightarrow \text{Option } (\text{Fin } n)$, $\varepsilon_i, \varphi_i: \text{Fin } n \rightarrow \mathbb{Z}$; `IsCrystal` (axioms 1–4) and `IsSeminormal`; `seminormal_eps_max` and `seminormal_phi_max` prove that it means $\varepsilon_i = \max\{k : \tilde{e}_i^k b \neq 0\}$ (and likewise for φ_i).
- Laurent polynomials as finite formal sums with coefficient-wise equality `PolyEq` (a sound and complete decision procedure is proved), multiplication, the S_3 -action, J , the orbit sum, the character `ch`.
- `schur`: Schur polynomials as sums over semistandard tableaux (all fillings of the shape with entries in $\{1, 2, 3\}$ are generated and filtered); the bialternant identities $s_\lambda a_\rho = a_{\lambda+\rho}$ for $\lambda \vdash 3$; `schur3_indep` (uniqueness of degree-3 Schur expansions).
- `WeylSym f q`: $q \cdot J(x^\rho) = J(x^\rho f)$; `schurCoeffW`: $[x^{\lambda+\rho}]J(x^\rho f)$.
- `B7_isCrystal`, `B7_seminormal`, `ch_B7_expands` ($\text{ch } B_7 = s_{210} - s_{111}$), `weylSym_B7`, `weyl_forced` (for *every* q with `WeylSym (ch B7) q` and every degree-3 expansion of q the coefficients are $(0, 1, -1)$), `orbit_forced`, `avg_forced`, `B7_counterexample`.
- Main theorems: `conjecture_00000003831_false`: $\neg \text{ClaimWeyl}$ and the variants `_abstract`, `_bialt`, `_orbit`, `_avg` for the other readings.

- Non-vacuity: `B100_ok` ($B(1, 0, 0)$ is a seminormal crystal with character s_{100}), `B210_ok` ($B(2, 1, 0)$ satisfies all hypotheses and has expansion s_{210} , nonnegative), `broken_rejected` (a datum violating axiom 1 is rejected).

Limitations: in Lean the Schur expansions quantified over are the degree-3 ones $as_{300} + bs_{210} + cs_{111}$ (sufficient because $\text{ch } B_7$ is homogeneous of degree 3; general uniqueness is Lemma 1). That the symmetrizer fixes symmetric polynomials is checked for $\text{ch } B_7$, not proved in general. Normality, Lemma 5, the $\mathfrak{gl}_2$ statement and minimality are in this report and `verify.py` only.

Python (`verify.py`, standard library). It checks the axioms by a different method (string lengths computed by walking), computes Schur polynomials both from tableaux and by exact division $a_{\lambda+\rho}/a_\rho$, computes the symmetrizer by exact polynomial division and via Demazure operators $\pi_1\pi_2\pi_1$, the orbit sum and average (with fractions), runs the $\mathfrak{gl}_2$ (up to 10 vertices) and $\mathfrak{gl}_3$ (up to 7 vertices) searches of Proposition 7, and sanity-checks the normal crystals $B(1, 0, 0)$, $B(2, 1, 0)$ and $B(1)^{\otimes 3}$ (tensor rule).

References

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